4. This Planck-Wheeler area is given by the formula $G\hbar/c^3$, where $G = 6.670 \times 10^{-8}$ dyne-cehtimeter^2/gram^2 is Newton's gravitation constant, $\hbar = 1.055 \times 1\ 0\text{-}^{27}$ erg-second is Planck's quantum mechanical constant, and $c = 2.998 \times 10^{10}$ centimeter/second is the speed of light. For related issues, see Footnote 2 in Chapter 13, Footnote 6 in Chapter 14, and the associated discussions in the text of those chapters.

5. The logarithm of 10^{1079} is 10^{79} (Bekenstein's conjectured entropy). Note that $10^{10^{79}}$ is a 1 with 10^{79} zeroes after it, that is, with nearly as many zeroes as there are atoms in the Universe.

6. Vorobyevskoye Shosse has since been renamed Kosygin Street, and its buildings have been renumbered. In the late 1980s Mikhail Gorbachev had a home at Number 10, several doors west of Zel'dovich.

7. See the last section of Chapter 1: "The Nature of Physical Law."

8. In technical language, the outer parts are in the "radiation zone" while the inner parts are in the "near zone."

9. Recall that photons and electromagnetic waves are different aspects of the same thing; see the discussion of wave/particle duality in Box 4.1.

10. This lack of interest was all the more remarkable because in the meantime, Charles Misner in America had shown that *real* waves (as opposed to Zel'dovich's vacuum fluctuations) can be amplified by a spinning hole in a manner analogous to Figure 12.1, and this amplification—to which Misner gave the name "superradiance"-—-was generating great interest.

11. The "almost" takes care of certain ambiguities in a procedure called "renormalization," by which one computes the net energy carried by vacuum fluctuations. These ambiguities, which were identified and codified by Robert Wald (a former student of Wheeler's), do not influence a black hole's evaporation, and they probably will not be resolved until the full quantum theory of gravity is in hand.

12. Some readers may already be familiar with these concepts in the context of matter and antimatter, for example, an electron (which is a particle of matter) and a positron (its antiparticle). Just as the electromagnetic field is the field aspect of a photon, so also there exists an electron field which is the field aspect of the electron and the positron. At locations where the electron field's vacuum fluctuations are momentarily large, a virtual electron and a virtual positron are likely to flash into existence, as a pair; when the field fluctuates down, the electron and positron are likely to annihilate each other and disappear. The photon is its own antiparticle, so virtual photons flash in and out of existence in pairs, and similarly for gravitons

13.Recall that, since mass and energy are totally convertible into each other, they are really just different names for the same concept.

14. The peculiar factor 0.10857 ... is actually 1/(4logel0), where loge10 = 2.30258 . . . results from my choice of "normalization" of the entropy; see Footnote 3 on page 423.